Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_random_replay_compression.
- Best TSPLIB holdout gap: phase6_random_replay_compression.
- Best synthetic holdout gap: phase6_failure_replay.

Run Metadata

- run_name: run_20260519_043037_s
- started_at_local: 2026-05-19 04:30:37
- finished_at_local: 2026-05-19 04:50:54
- duration_hhmm: 00:20
- duration_seconds: 1217.138
- seed_offset: 18000
- replicate_label: s
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.0	0.840932	0.56	-0.029616
phase6_random_replay	random	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.19457	0.901541	0.56	-0.027625
phase6_failure_replay	failure	score_only	False	0.109966	0.0	0.069978	0.171206	0.273059	0.214199	0.815934	0.76	0.061169

Condition	**Replay**	**Selection**	**Compression**	**Final TSPLIB Gap**	**Final Synthetic Gap**	**Final Transfer Gap**	**Archive Diversity**	**Archive Hardness**	**Failure Concentration**	**Mean Accepted Novelty**	**Mean Complexity**	**Adaptation Efficiency**
phase6_random_replay_compression	random	novelty_gate	True	0.109154	0.0	0.069462	0.236772	0.15889	0.19457	0.690544	0.76	0.043613
phase6_stratified_random_replay	stratified_random	score_only	False	0.185243	0.001628	0.118474	0.236772	0.193893	0.221162	0.654646	0.76	0.054327
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.16851	0.0	0.107234	0.236772	0.165675	0.130039	0.756331	0.76	0.019698
phase6_residual_failure_replay	residual_failure	score_only	False	0.192759	0.0	0.122665	0.113626	0.251656	0.256191	0.0	0.76	0.0
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.213387	0.064916	0.159398	0.171206	0.348017	0.17024	0.842788	0.56	-0.029551
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.136485	0.015447	0.092471	0.171206	0.244383	0.171578	0.0	0.76	0.0

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.840932, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency -0.029616 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.404343, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.003025}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.133545},

{"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.328729, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 8.4e-05}]

phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.901541, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.134046, mean selected residual gap 0.03732.
- Adaptation efficiency -0.027625 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.40442, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.003102}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.134817}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.328131, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.000682}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.109966, synthetic 0.0, combined 0.069978.
- Accepted-epoch count 2, mean accepted code novelty 0.815934, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.273059, size bias 0.120322, and failure concentration 0.214199.
- Replay selection diversity 0.192353, mean selected expected gap 0.321931, mean selected residual gap -0.012243.
- Adaptation efficiency 0.061169 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.109966 across 7 instances; family means: ch=0.126975, kroD=0.072697, pcb=0.242388, pr=0.099955, rd=0.02225, st=0.078519.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3648, "expected_gap": 0.406824, "family": "a", "name": "a280", "optimality_gap": 0.414502, "residual_gap": 0.007678}, {"best_known_cost": 629, "cost": 858, "expected_gap": 0.25469, "family": "eil", "name": "eil101", "optimality_gap": 0.36407, "residual_gap": 0.10938}, {"best_known_cost": 14379, "cost": 17626, "expected_gap": 0.329634, "family": "lin", "name": "lin105", "optimality_gap": 0.225815, "residual_gap": -0.103819}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.109154, synthetic 0.0, combined 0.069462.

- Accepted-epoch count 3, mean accepted code novelty 0.690544, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.15889, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132412, mean selected residual gap -0.017766.
- Adaptation efficiency 0.043613 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.109154 across 7 instances; family means: ch=0.126737, kroD=0.112943, pcb=0.226062, pr=0.091976, rd=0.036662, st=0.042963.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3179, "expected_gap": 0.404498, "family": "a", "name": "a280", "optimality_gap": 0.232648, "residual_gap": -0.17185}, {"best_known_cost": 14379, "cost": 16217, "expected_gap": 0.322762, "family": "lin", "name": "lin105", "optimality_gap": 0.127825, "residual_gap": -0.194937}, {"best_known_cost": 7542, "cost": 8217, "expected_gap": 0.240918, "family": "berlin", "name": "berlin52", "optimality_gap": 0.089499, "residual_gap": -0.151419}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.185243, synthetic 0.001628, combined 0.118474.
- Accepted-epoch count 2, mean accepted code novelty 0.654646, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.193893, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.236305, mean selected residual gap -0.000366.
- Adaptation efficiency 0.054327 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.185243 across 7 instances; family means: ch=0.135823, kroD=0.245093, pcb=0.259798, pr=0.226833, rd=0.2, st=0.093333.
- Panel synthetic_holdout mean gap 0.001628 across 4 instances; family means: clustered_gaussian=0.006511, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3475, "expected_gap": 0.405661, "family": "a", "name": "a280", "optimality_gap": 0.347421, "residual_gap": -0.05824}, {"best_known_cost": 7542, "cost": 10022, "expected_gap": 0.236622, "family": "berlin", "name": "berlin52", "optimality_gap": 0.328825, "residual_gap": 0.092203}, {"best_known_cost": 629, "cost": 741, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.17806, "residual_gap": -0.07822}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.16851, synthetic 0.0, combined 0.107234.
- Accepted-epoch count 2, mean accepted code novelty 0.756331, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.165675, size bias 8.3e-05, and failure concentration 0.130039.
- Replay selection diversity 0.270615, mean selected expected gap 0.151903, mean selected residual gap -0.015067.

- Adaptation efficiency 0.019698 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.16851 across 7 instances; family means: ch=0.159265, kroD=0.150606, pcb=0.213675, pr=0.318633, rd=0.130721, st=0.047407.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3295, "expected_gap": 0.404188, "family": "a", "name": "a280", "optimality_gap": 0.277627, "residual_gap": -0.126561}, {"best_known_cost": 14379, "cost": 18091, "expected_gap": 0.329244, "family": "lin", "name": "lin105", "optimality_gap": 0.258154, "residual_gap": -0.07109}, {"best_known_cost": 629, "cost": 675, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.073132, "residual_gap": -0.180922}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.192759, synthetic 0.0, combined 0.122665.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive residual_archive with mean diversity 0.113626, mean hardness 0.251656, size bias 0.072026, and failure concentration 0.256191.
- Replay selection diversity 0.151242, mean selected expected gap 0.284426, mean selected residual gap 0.051316.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.192759 across 7 instances; family means: ch=0.191087, kroD=0.341035, pcb=0.195144, pr=0.235736, rd=0.170038, st=0.025185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3544, "expected_gap": 0.405196, "family": "a", "name": "a280", "optimality_gap": 0.374176, "residual_gap": -0.03102}, {"best_known_cost": 629, "cost": 848, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.348172, "residual_gap": 0.086169}, {"best_known_cost": 14379, "cost": 18820, "expected_gap": 0.328841, "family": "lin", "name": "lin105", "optimality_gap": 0.308853, "residual_gap": -0.019988}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.842788, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.17024.
- Replay selection diversity 0.199284, mean selected expected gap 0.301037, mean selected residual gap 0.048952.
- Adaptation efficiency -0.029551 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.403955, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.002637}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.124006}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.328006, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.000807}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.136485, synthetic 0.015447, combined 0.092471.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.244383, size bias 0.120322, and failure concentration 0.171578.
- Replay selection diversity 0.197858, mean selected expected gap 0.3103, mean selected residual gap -0.054684.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.136485 across 7 instances; family means: ch=0.136816, kroD=0.200855, pcb=0.192918, pr=0.133378, rd=0.107206, st=0.047407.
- Panel synthetic_holdout mean gap 0.015447 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.061786, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3450, "expected_gap": 0.407212, "family": "a", "name": "a280", "optimality_gap": 0.337728, "residual_gap": -0.069484}, {"best_known_cost": 7542, "cost": 9644, "expected_gap": 0.233068, "family": "berlin", "name": "berlin52", "optimality_gap": 0.278706, "residual_gap": 0.045638}, {"best_known_cost": 14379, "cost": 17366, "expected_gap": 0.326657, "family": "lin", "name": "lin105", "optimality_gap": 0.207734, "residual_gap": -0.118923}]

Judge Appendix

Summary of Replay-Aware TSP Benchmark Suite Results

Overview

- **Best holdout condition:** phase6_random_replay_compression
- **Best synthetic condition:** phase6_failure_replay
- **Best transfer condition:** phase6_random_replay_compression
- **Total conditions analyzed:** 9

Transfer Performance (Key Metrics)

Condition	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap
phase6_no_replay	0.213387	0.064916	0.159398

Condition	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap
phase6_random_replay	0.213387	0.064916	0.159398
phase6_failure_replay	**0.109966**	**0.000000**	**0.069978**
phase6_random_replay_compression	**0.109154**	**0.000000**	**0.069462**
phase6_stratified_random_replay	0.185243	0.001628	0.118474
phase6_diversity_weighted_replay	0.168510	0.000000	0.107234
phase6_residual_failure_replay	0.192759	0.000000	0.122665
phase6_diversity_failure_replay	0.213387	0.064916	0.159398
phase6_diversity_failure_replay_compression	0.136485	0.015447	0.092471

- **Inference:**
- phase6_failure_replay and phase6_random_replay_compression achieve the lowest TSPLIB and synthetic holdout gaps, indicating superior transfer performance.
- The no replay baseline and simple random replay achieve higher gaps, showing less transfer effectiveness.
- Stratified random, diversity-weighted, and residual failure replays show intermediate transfer gaps.
- Diversity failure replays show weaker transfer despite some replay.
- Compression in replay can improve transfer gap significantly (phase6_random_replay_compression vs phase6_random_replay), even with some cost to code novelty.

Replay Archive Characteristics and Mechanism Insights

Condition	Replay Mode	Compression	Archive Diversity	Archive Hardness	Archive Size Bias	Failure Concentration	Selection Diversity	Code Novelty Mean	Complexity Mean
phase6_no_replay	none	No	0.240179	0.212361	~0.0	0.0	0.0	0.840932	0.56
phase6_random_replay	random	No	0.240179	0.212361	~0.0	0.19457	0.262174	0.901541	0.56
phase6_failure_replay	failure	No	0.174718	0.334281	0.124979	0.214199	0.192353	0.815934	0.76
phase6_random_replay_compression	random	Yes	0.240179	0.166844	~0.0	0.19457	0.262174	0.690544	0.76
phase6_stratified_random_replay	stratified_random	No	0.240179	0.204653	~0.0	0.221162	0.187367	0.654646	0.76
phase6_diversity_weighted_replay	diversity_weighted	No	0.240179	0.184688	~0.0	0.130039	0.270615	0.756331	0.76
phase6_residual_failure_replay	residual_failure	No	0.167691	0.348023	0.089671	0.256191	0.151242	0.0	0.76

Condition	Replay Mode	Compression	Archive Diversity	Archive Hardness	Archive Size Bias	Failure Concentration	Selection Diversity	Code Novelty Mean	Complexity Mean
phase6_diversity_failure_replay	diversity_failure	No	0.174718	0.34902	0.124979	0.17024	0.199284	0.842788	0.56
phase6_diversity_failure_replay_compression	diversity_failure	Yes	0.174718	0.319545	0.124979	0.171578	0.197858	0.0	0.76

- **Observations:**
- Replay diversification mechanisms (random, stratified random, diversity-weighted) maintain higher archive descriptor diversity and balanced hardness.
- Failure replays (failure, residual_failure, diversity_failure) result in higher archive hardness and some positive archive size bias.
- Compression pressures decrease average code novelty but can improve transfer gaps (see phase6_random_replay_compression).
- Residual failure replays exhibit highest failure concentration and lowest code novelty, with zero mean code novelty in residual and diversity failure with compression.
- Selection diversity highest in random and diversity-weighted replays, lowest when replay mode is none or residual failure.
- Elite archive sizes are larger with replay than no replay, possibly indicating more robust elite selection.

Optimality Gap and Failure Concentration Summary

- **Best TSPLIB gaps (~0.11) and transfer gaps (~0.07):** Seen in phase6_failure_replay and phase6_random_replay_compression.
- **Baseline (no replay):** TSPLIB gap ~0.21 with no replay failure concentration and no replay selection diversity.
- Failure replay modes help reduce gaps but tend to concentrate failures more (0.21-0.26 failure concentration).
- Compression-aware replay (phase6_random_replay_compression) achieves low transfer and TSPLIB gaps with moderate failure concentration (0.19 to 0.25) and balanced diversity.

Code Novelty and Transfer Relation

- phase6_random_replay_compression reduces code novelty mean (~0.69) compared to random replay (~0.90) but improves transfer metrics, indicating **code novelty decrease concurrent with transfer improvement**. This suggests replay compression drives more general transfer with less code diversity.
- Residual failure and diversity failure replays with compression show zero mean code novelty, suggesting replay pressure impacts code novelty.
- High code novelty alone (e.g., random replay) does not guarantee best transfer.

Replay Mode Distinctions

- **No replay**: No replay, no failure concentration, zero selection diversity, high code novelty but poor transfer gaps.
- **Random replay**: Broad, random replays increase selection diversity with moderate failure concentration and no compression pressure, similar transfer to no replay.
- **Failure replay**: Replay conditioned on failure cases increases archive hardness and transfer, reduces synthetic gap to 0.0, but concentrates failures moderately (~0.21).
- **Random replay compression**: Broad coverage replay with compression pressure reduces archive hardness, reduces transfer gap most significantly, with moderate failure concentration and reduced code novelty.
- **Stratified random replay**: More structured replay slightly improves over random replay, but not as good as failure-based or compression replay.
- **Diversity-weighted replay**: Maintains high archive diversity and selection diversity with moderate hardness, resulting in intermediate transfer gains.
- **Residual failure replay**: Replay samples residual failures only, with highest failure concentration (~0.25), zero mean code novelty, and moderate TSPLIB gap (worse than failure replay).
- **Diversity failure replay**: Combines failure replay with diversity weighting, but results in no transfer improvement over baseline, likely due to complexity or failure concentration effects.
- **Diversity failure replay compression**: Adds compression to diversity failure replay, slightly better than diversity failure alone but still worse than failure replay.

Summary Conclusions

- Failure-based replay (phase6_failure_replay) and random replay with compression (phase6_random_replay_compression) achieve **best overall transfer to holdout TSPLIB and synthetic instances**, halving mean gaps compared to no-replay baseline.
- Compression-aware replay reduces code novelty but enhances transfer, indicating code novelty decrease co-occurs with transfer improvement here.
- Random and diversity-weighted replays increase code novelty and selection diversity but do not improve transfer as effectively as failure replay.
- Residual failure replay has concentrated failures in archive, zero code novelty, limiting transfer benefits.
- Diversity failure replay conditions produce poor transfer performance, suggesting failure concentration and replay complexity hinder benefit.
- Archive diversity tracks with replay mode designed for broad coverage and diversity weighting, while failure replay trades off diversity for hardness and transfer gains.

Key Recommendation

Employ failure replay and compression-aware broad coverage replay protocols for most effective transfer with balanced archive diversity, moderate failure concentration, and good final complexity. Avoid residual-only failure and diversity failure replays as they show weak transfer despite potential replay mechanism sophistication.

End of Analysis